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Reg No.: _____

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APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

B.Tech S1 (Special Improvement) Examinations January 2021 (2019 scheme)

Course Code: PHT100

**Course Name: ENGINEERING PHYSICS A
(2019-Scheme)**

Max. Marks: 100

Duration: 3 Hours

PART A

Answer all questions, each carries 3 marks.

Marks

- | | | |
|----|---|-----|
| 1 | Distinguish between free oscillation and damped oscillation. | (3) |
| 2 | Define frequency, wavelength and wave velocity of a wave. | (3) |
| 3 | Write a short note on colours of thin films. | (3) |
| 4 | What is meant by the phenomenon of diffraction? Why diffraction of light is not evident in daily experience as that of sound? | (3) |
| 5 | What are matter waves? Derive the expression for de-Broglie wavelength. | (3) |
| 6 | Explain the effect of increased surface to volume ratio in nanomaterials. | (3) |
| 7 | State Faradays laws of electromagnetic induction and Lenz's law. | (3) |
| 8 | Give the physical significance of curl. | (3) |
| 9 | What is critical magnetic field? How is it related to temperature of superconductor? | (3) |
| 10 | What is a light emitting diode? Give its working principle. | (3) |

PART B

Answer one full question from each module, each question carries 14 marks

Module-I

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|----|---|------|
| 11 | a) Write down the differential equation of a forced harmonic oscillator and obtain its solution. Derive the expressions for amplitude and phase difference. | (10) |
| | b) A transverse wave on a stretched string is described by
$y(x,t)=5 \sin(25t+0.016x+\pi/2)$ where x and y are in cm and t is in second.
Obtain (1) Speed (2) Amplitude (3) Frequency and (4) Initial phase of the wave | (4) |
| 12 | a) Derive an expression for the velocity of transverse waves in a stretched string and state the laws of transverse vibrations. | (10) |

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- b) A piece of wire **60 cm** long and mass **1.2 g**. is stretched by a load of **3 kg**. Find (4)
the frequency of the second harmonic.

Module-II

- 13 a) Starting from the expression of radius of n th dark ring in Newton's rings (10)
pattern, describe an experiment to determine the refractive index of a transparent liquid.
- b) Two optically plane glass plates of length **0.1m** are placed one over the other (4)
with a thin wire at one end, separating the two. The fringes formed with light of wavelength **589.3 nm** are of width **3mm**. Calculate radius of the wire.
- 14 a) Derive grating equation for a plane transmission grating. Explain resolving (10)
power and dispersive power of grating with expressions.
- b) When a diffraction grating is used at normal incidence, it is found that the (4)
image at **30°** consists of a yellow line of wavelength **5750 Å** of the **n th** order spectrum is superimposed on a blue line of wavelength **4600 Å** of order **$(n+1)$** . Calculate the number of lines per unit length of grating.

Module-III

- 15 a) State and explain uncertainty principle. Write the three forms of uncertainty (10)
relations. How this principle is used to prove the absence of electron in the nucleus? Given $m_e = 9.1 \times 10^{-31} \text{ kg}$; $h = 6.625 \times 10^{-34} \text{ Js}$
- b) For an electron in a one dimensional box of width **1Å**, calculate the first three (4)
energy levels in **electron volt**.
- 16 a) Why do nanomaterials exhibit properties different from those of their classical (10)
counter parts? Explain the electrical and mechanical properties of nanomaterials.
- b) Mention any four applications of nanotechnology. (4)

Module-IV

- 17 a) Compare the properties of paramagnetic, diamagnetic and ferromagnetic (10)
materials.
- b) Find the relative permeability of a ferromagnetic material if a field strength of (4)
200 A/m produces a magnetization of **3100 A/m**.
- 18 a) Starting from Maxwell's equations show that electromagnetic waves are (10)
existing in free space and find an expression for velocity.

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- b) Calculate the value of Poynting's vector at the surface of the sun if the power radiated by sun is 3.8×10^{26} Watts and its radius is 7×10^8 m. (4)

Module-V

- 19 a) Write a note on high temperature superconductors. Distinguish between Type I and Type II superconductors with appropriate diagrams and examples. (10)
- b) Mention any four applications of superconductivity. (4)
- 20 a) Draw the block diagram of optical fibre communication system and explain its various functional blocks. Mention the advantages of optical fibres over conventional transmission lines. (8)
- b) What are sensors? Explain the working of intensity modulated sensor. (6)
